

tf.compat.v1.nn.separable_conv2d Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/separable_conv2d

2-D convolution with separable filters.

Function Signatures

```
tf.compat.v1.nn.separable_conv2d( input, depthwise_filter,  
pointwise_filter, strides, padding, rate=None, name=None,  
data_format=None, dilations=None )
```

Code Examples

```
tf.compat.v1.nn.separable_conv2d(  
    input,  
    depthwise_filter,  
    pointwise_filter,  
    strides,  
    padding,  
    rate=None,  
    name=None,  
    data_format=None,  
    dilations=None  
)
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    input,  
    depthwise_filter,  
    pointwise_filter,  
    strides,  
    padding,  
    rate=None,  
    name=None,  
    data_format=None,  
    dilations=None  
)
```

```
output[b, i, j, k] = sum_{di, dj, q, r}  
input[b, strides[1] * i + di, strides[2] * j + dj, q] *  
depthwise_filter[di, dj, q, r] *  
pointwise_filter[0, 0, q * channel_multiplier + r, k]
```

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output[b, i, j, k] = sum_{di, dj, q, r}  
input[b, strides[1] * i + di, strides[2] * j + dj, q] *  
depthwise_filter[di, dj, q, r] *  
pointwise_filter[0, 0, q * channel_multiplier + r, k]
```

Documentation

2-D convolution with separable filters.

Performs a depthwise convolution that acts separately on channels followed by a pointwise convolution that mixes channels. Note that this is separability between dimensions [1, 2] and 3, not spatial separability between dimensions 1 and 2.

In detail, with the default NHWC format,

`strides` controls the strides for the depthwise convolution only, since the pointwise convolution has implicit strides of [1, 1, 1, 1]. Must have `strides[0] = strides[3] = 1`. For the most common case of the same horizontal and vertical strides, `strides = [1, stride, stride, 1]`.

If any value in `rate` is greater than 1, we perform atrous depthwise convolution, in which case all values in the `strides` tensor must be equal to 1.

Returns